National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	**Computer Networks**	**Course Code:**	**CS 3001**
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Duration:	**15 minutes**	**Total Marks:**	**15**
	Paper Date:		**Section**	**6B1**
	Exam Type:	**Quiz 2 - Chapter 2**	**Page(s):**	**2**

Student Name **Roll No.** **Section:**

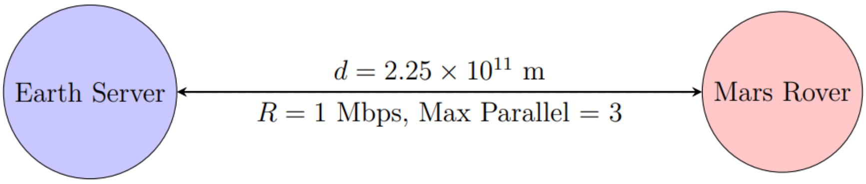
Q1. Carefully read the Question before attempting: **[15 marks]**

An automated rover on Mars needs to download a firmware update from a server on Earth. The update consists of one base HTML index file (size L_b = 2 KB) and 8 referenced binary objects (each size L_o = 100 KB). The link characteristics are:

- Distance Earth-Mars: d = 2.25×10^{11} meters.
- Transmission Rate: R = 1 Mbps.

The rover opens parallel TCP connections but is limited to a maximum of K = 3 parallel connections due to memory constraints. Calculate the total time elapsed from the initial connection request until the last object is fully received. Assume:

- Short control packets (SYN, ACK, HTTP GET) have negligible transmission time compared to propagation delay.
- We are using Non-Persistent HTTP over the parallel connections. No DNS lookup is required (IP is hardcoded). Speed of light c = 3×10^8 m/s.



Solution:

1. **Calculations:**

$$d_{prop} = \frac{2.25 \times 10^{11}}{3 \times 10^8} = 750 \text{ s} \implies RTT = 2 \times d_{prop} = 1500 \text{ s}$$

$$d_{trans(base)} = \frac{2 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.016 \text{ s}, \quad d_{trans(obj)} = \frac{100 \text{ KB} \times 8}{1 \text{ Mbps}} = 0.8 \text{ s}$$

2. **Step 1: Base File Download** (Non-Persistent: 2 RTT + Trans)

$$T_{base} = 2 \times RTT + d_{trans(base)} = 3000 + 0.016 = 3000.016 \text{ s}$$

3. **Step 2: 8 Objects with K=3 Parallel Connections** We download objects in batches. Since $K = 3$, we need $\lceil 8/3 \rceil = 3$ batches. Note: Bandwidth is shared, so transmission time for 3 objects is $3 \times d_{trans(obj)}$.
   - **Batch 1 (Obj 1-3):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4$ s
   - **Batch 2 (Obj 4-6):** $2RTT + 3 \times 0.8 = 3000 + 2.4 = 3002.4$ s
   - **Batch 3 (Obj 7-8):** $2RTT + 2 \times 0.8 = 3000 + 1.6 = 3001.6$ s

4. **Total Time:**

$$T_{total} = 3000.016 + 3002.4 + 3002.4 + 3001.6 = \mathbf{12006.416 \text{ s}}$$